

**Definition:** A function  $f$  is **continuous at a number  $a$**  if  $\lim_{x \rightarrow a} f(x) = f(a)$ .

**Theorem:** If  $f$  and  $g$  are continuous at  $a$  and  $c$  is constant, then the following functions are also continuous at  $a$ :

1.  $f + g$
2.  $f - g$
3.  $cf$
4.  $fg$
5.  $\frac{f}{g}$  if  $g(a) \neq 0$

**Theorem:** The following types of functions are continuous at every number in their domains.

- Polynomial Functions
- Rational Functions
- Root Functions
- Trigonometric Functions
- Inverse Trigonometric Functions
- Exponential Functions
- Logarithmic Functions

**Theorem:** If  $f$  is continuous at  $b$  and  $\lim_{x \rightarrow a} g(x) = b$ , then

$$\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right) = f(b).$$

**Theorem (The Intermediate Value Theorem):**

Suppose that  $f$  is continuous on the closed interval  $[a, b]$  and let  $N$  be any number between  $f(a)$  and  $f(b)$  where  $f(a) \neq f(b)$ . Then, there exists a number  $c$  in  $(a, b)$  such that  $f(c) = N$ .